

A Local Agent, Knowledge, and Voice Architecture

An anonymized briefing on how a workstation, local agent host, durable knowledge, training workflows, and voice interface work together.



Workstation

Canvas • coding • messaging • voice



Local agent host

Hermes • models • tools • training



Private durability

GBrain • authored sources • Git

ARCHITECTURE AUDIT • JULY 2026

Hermes = runtime • GBrain = durable knowledge • TARS = voice/persona

The four-layer mental model

1. Interaction

Where a person asks, observes, and approves.

Canvas
Messaging
Coding sessions
Voice

2. Agent

Where requests are authenticated, scoped, reasoned about, and executed.

Gateway
Hermes sessions
Local models
Approved tools

3. Knowledge

Where current evidence is reconciled with durable context.

Live run state
Scoped GBrain
Training receipts
Provenance

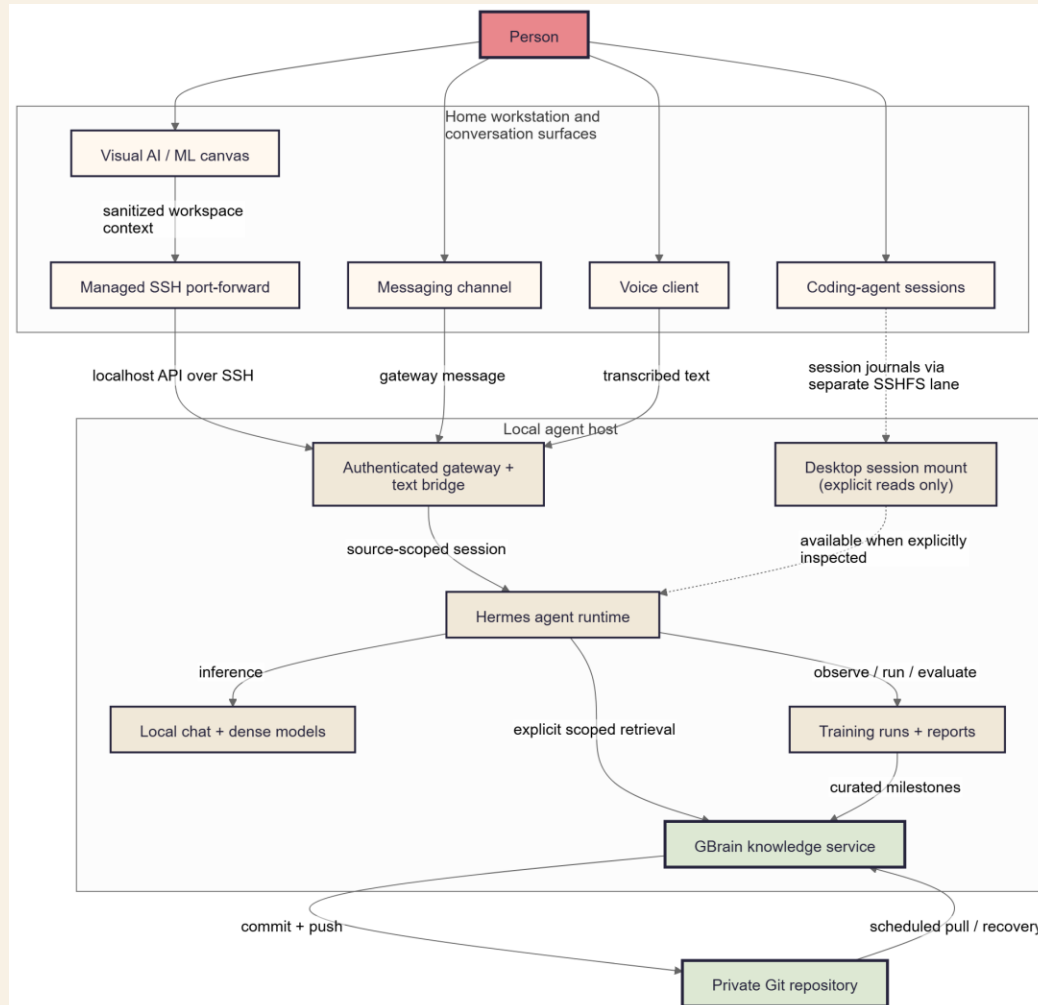
4. Durability

Where authored knowledge can be backed up and restored.

Markdown sources
Local index
Private Git
Recovery

The workstation controls. The local host executes. GBrain remembers.

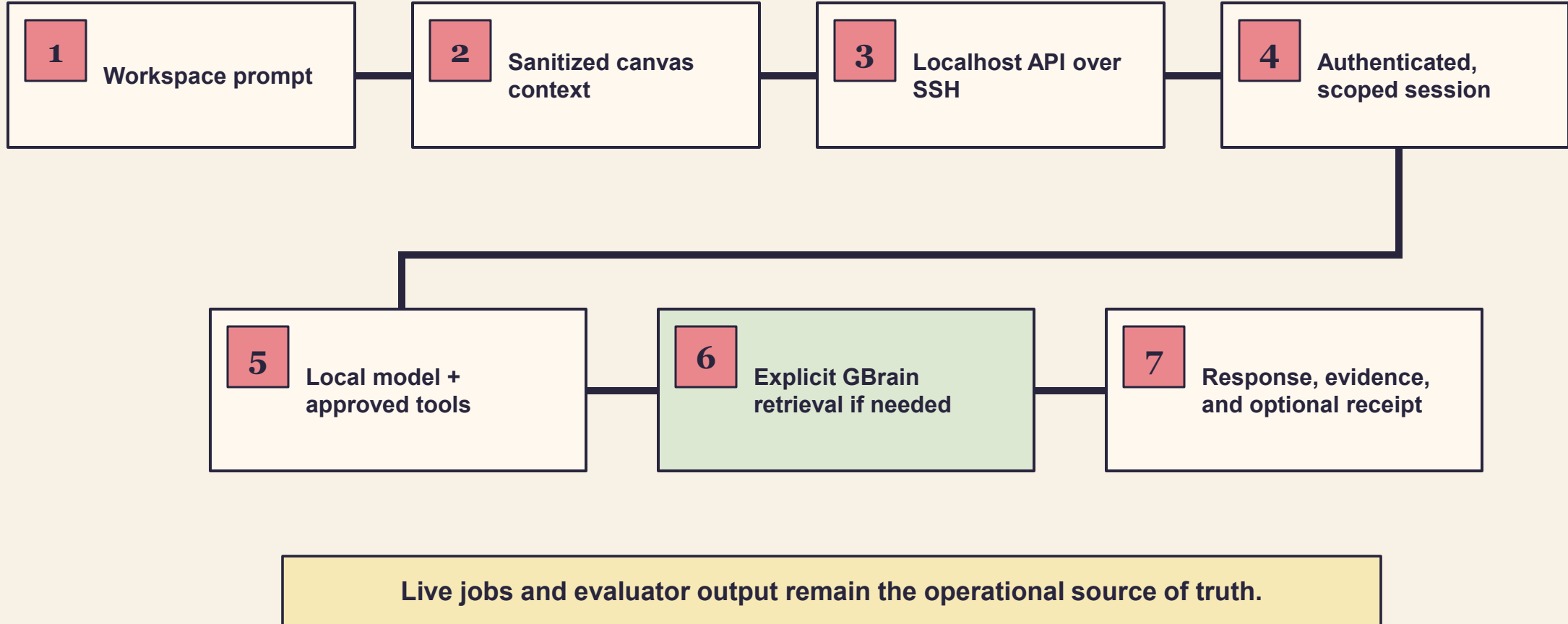
One system, several deliberately separate lanes



WHAT CROSSES THE BOUNDARY

- Canvas: sanitized workspace context over a managed SSH port-forward
- Messaging: a source-scoped gateway message
- Voice: transcribed text over a trusted local network
- Coding sessions: explicit reads through a separate filesystem lane
- Durability: authored GBrain sources commit and push outside the request path

What happens during one canvas request



Context is scoped—not magically shared

Automatic

- Sanitized canvas context with each prompt
- Stable source-specific session identity
- Gateway authentication and transport
- Configured local model and approved tools

Explicit

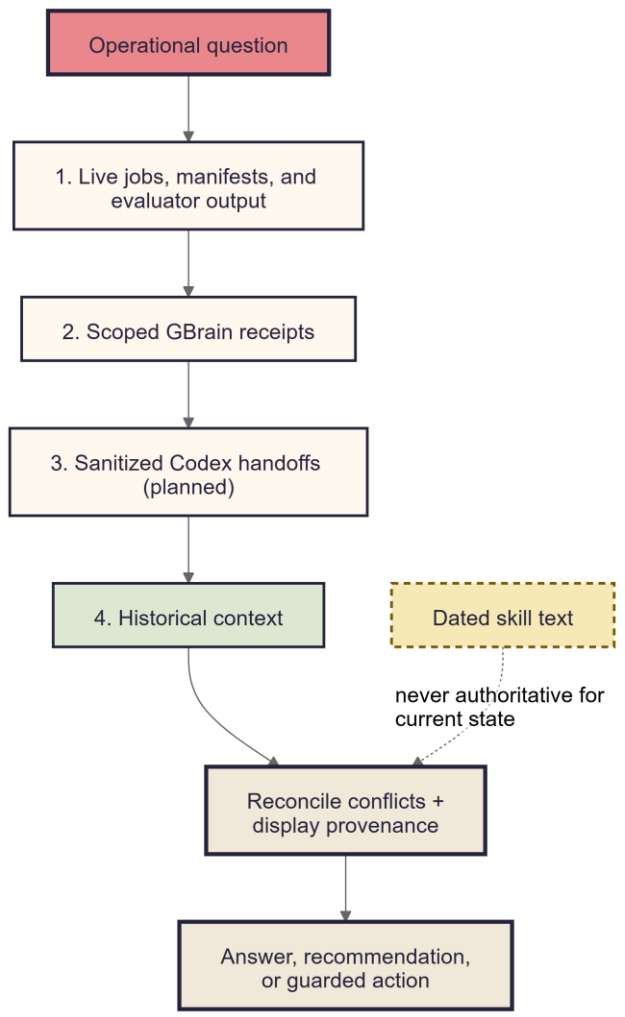
- Scoped GBrain retrieval
- Coding-session journal inspection
- Training, evaluation, and promotion workflows
- Cross-surface provenance reconciliation

Not implied

- One giant transcript across surfaces
- Automatic indexing of mounted directories
- Copying all raw logs into GBrain
- Old skill text overriding live state

Availability is not ingestion. History is not current state.

Fresh evidence first. Durable context second.



- 1

Live operations
Jobs, manifests, evaluator output, and artifacts answer what is happening now.
- 2

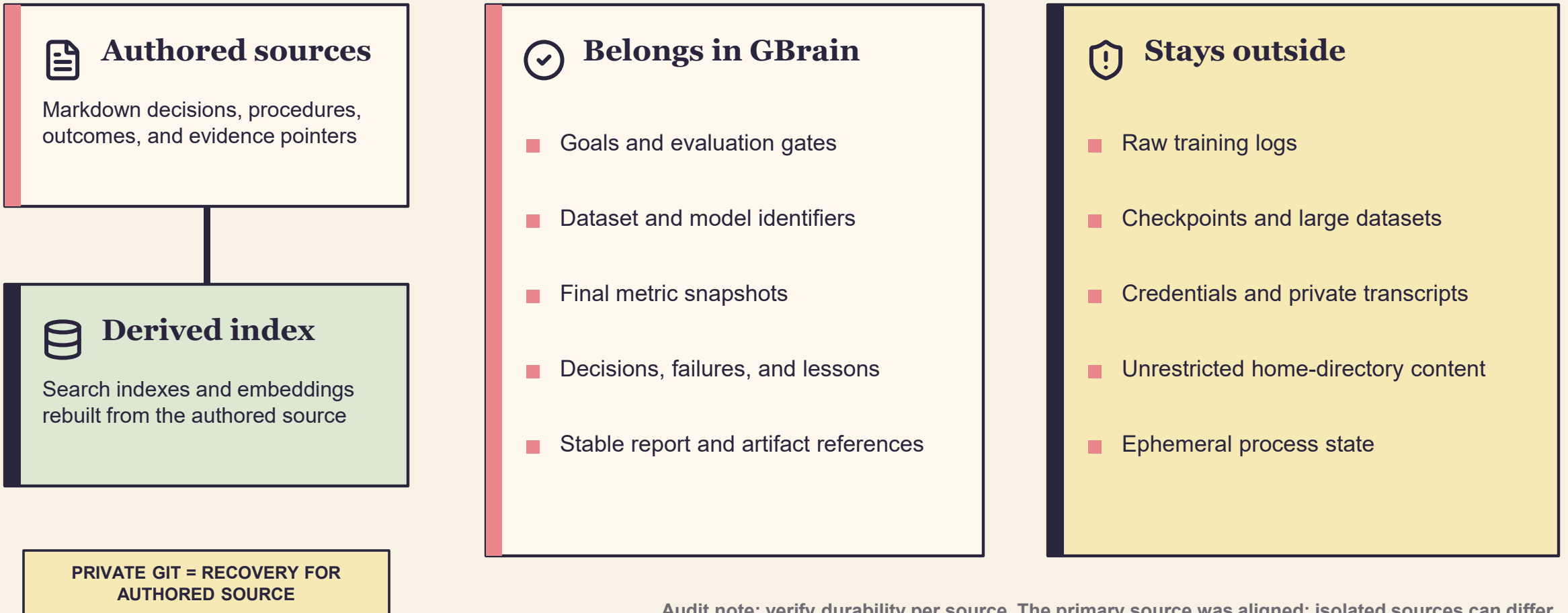
Scoped GBrain receipts
Goals, prior decisions, and verified outcomes explain why the system is here.
- 3

Sanitized handoffs
Recent coding context becomes useful after it is explicitly published and bounded.
- 4

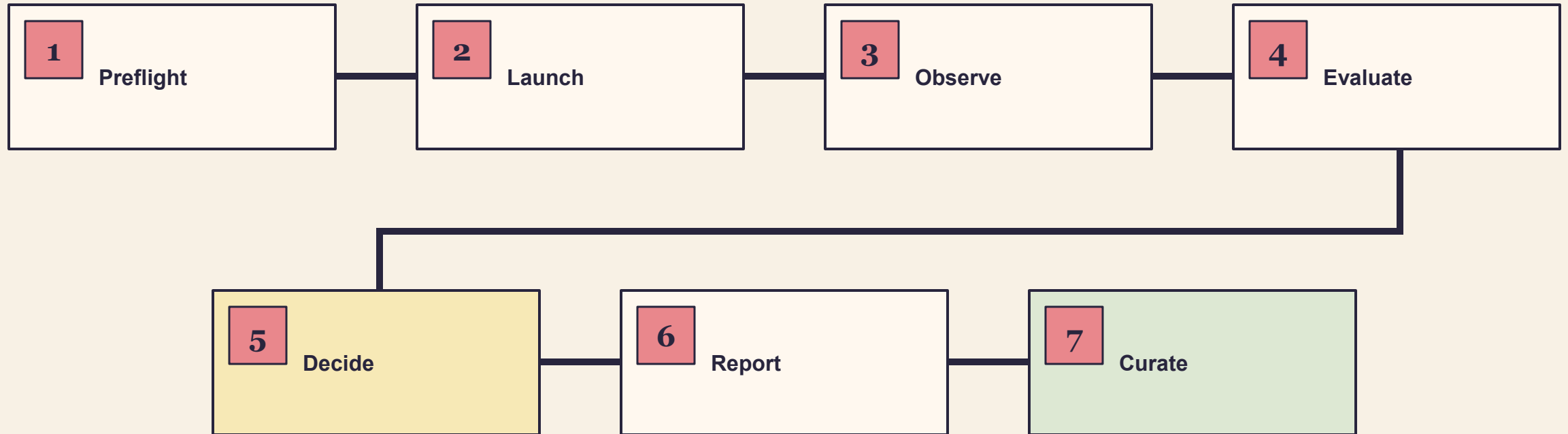
Historical context
Older sessions support an answer; they do not become authority.

Dated skills never override current evidence.

GBrain stores durable receipts—not the whole machine



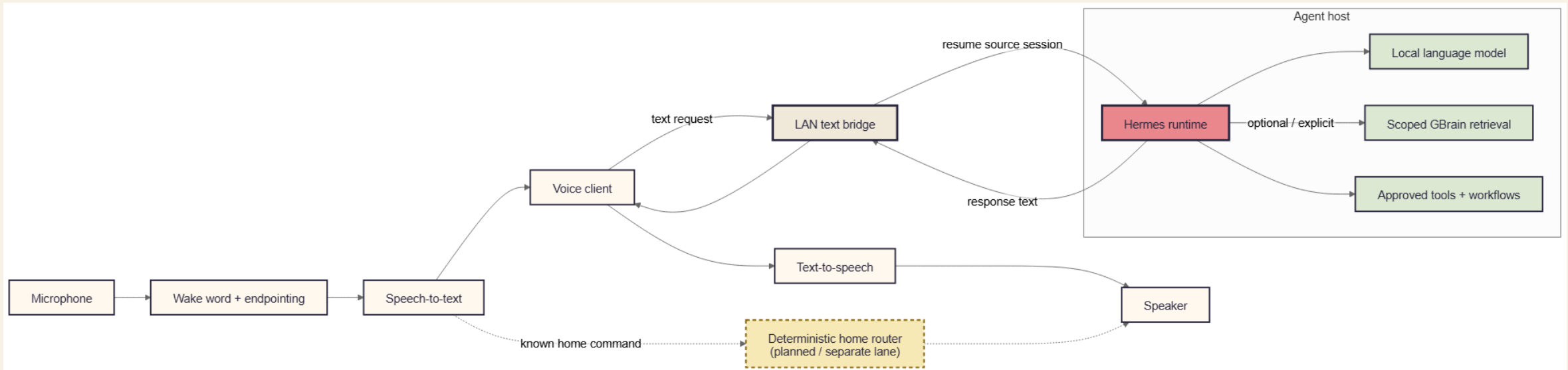
The agent operates a task-general training loop



Embedding fine-tuning is one profile—not the identity of the platform.

The same operator pattern can support SFT, preference optimization, RL workflows, distillation, adapters, and future full-model training.

TARS is the voice/persona layer around the same agent



Local audio

Wake word • endpointing • speech-to-text
• text-to-speech

Same agent runtime

Source-scoped Hermes session with
optional tools and GBrain

Deterministic lane

Planned path for known home commands
that should stay fast and predictable

Working now, limited now, and planned next

Implementation status snapshot

SANITIZED ARCHITECTURE AUDIT • JULY 2026

WORKING NOW

- Authenticated text gateway
- Canvas workspace context
- Source-scoped messaging sessions
- Local model inference and approved tools
- Explicit, scoped GBrain retrieval
- Primary knowledge source aligned with private Git
- Training receipts and report references

WORKING WITH LIMITS

- Desktop journals are available only when explicitly inspected
- Voice text bridge is active
- Warm voice persona mode is unavailable
- Live messaging-room audio awaits a media dependency
- Edge microphone-to-speaker loop is not yet verified end to end

PLANNED / HARDEN NEXT

- Publish sanitized coding-session handoffs
- Add a deterministic local command lane
- Replace broad desktop mount with a narrow read-only bridge
- Verify voice-user authorization
- Continuously test push freshness and source-specific recovery

TARS Voice Persona

THESE SETTINGS SHAPE PI
VOICE TURNS
PERSONALITY

warm: All connection attempts failed

VOICE

WORKSPACE

VOICE HARDENING CHECKLIST

- ☐ Restore and test warm persona service
- ☐ Install live-room media dependency
- ☐ Run microphone-to-speaker trace
- ☐ Verify voice-user authorization

Trust boundaries are part of the product

Canvas → agent host

Current: authenticated API through a managed SSH port-forward.

Harden: rotate credentials and preserve redaction tests.

Voice → text bridge

Current: trusted-LAN text request.

Harden: explicit caller authorization or an authenticated tunnel.

Workstation journals

Current: separate filesystem bridge with explicit reads.

Harden: narrow, read-only handoff folder instead of a broad writable mount.

GBrain → private Git

Current: authored-source backup and scheduled recovery.

Harden: monitor push failures and verify policy for every isolated source.

A LAN listener is not automatically safe. A mounted folder is not automatically knowledge.

How to describe the system

🔊 30-second version

“The interface lives on a normal workstation, while the agent, models, training tools, and durable knowledge run on a dedicated local host. Canvas, messaging, and voice connect through authenticated, source-scoped sessions. The agent uses current job evidence first and queries GBrain only when durable history is useful. GBrain stores curated outcomes backed by private Git. Voice is another input and output layer around the same agent—not a separate brain.”

☑️ Technical walkthrough

- Name the interaction surface and session scope
- Explain the authenticated transport
- Separate automatic context from explicit retrieval
- Identify the current source of truth
- Show where reports, receipts, and recovery live

Do not claim shared transcripts, automatic ingestion, universal Git backup, or production-ready voice without evidence.